Excluded Women: The Fall of Female Labour Force Participation in Post-Industrial Revolution England*

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Abstract

We investigate the decline in female labour force participation in England between 1851 and 1911 using newly digitized census microdata. We document three new empirical findings. First, the overall female participation rate fell slightly, driven by a sharp drop among married women. Second, synthetic cohort analysis shows that this decline was driven primarily by younger married women choosing not to enter the workforce, rather than older women exiting earlier. Third, we show that women's employment was concentrated in a narrow set of sectors, primarily textiles and domestic service. We develop a theoretical model in which rising institutional and societal barriers limit married women's participation, providing a framework to interpret these long-run changes.

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1 Introduction

There are opposing views on the role that the Industrial Revolution had on female labor force participation. Mokyr (2010) asserts that "the Industrial Revolution was responsible for a marginalization of female labor and the gradual relegation of women to the position of homemaker." Essentially, he argues that the Industrial Revolution contributed to the exclusion of women from the labor market. Conversely, Burnette (2008) claims the opposite, suggesting that the Industrial Revolution eroded the male market power needed to restrict female employment, thereby improving the status of women. Analysis of women's labor force participation in the United Sates from the late nineteenth and early twentieth centuries, as documented by scholars such as Goldin (1989), reveals a gradual inclusion of women into the labor force. However, little is known about women's status in the labor force in the post-industrial revolution period in England. Understanding the economic status of women in this period can help us understand long-run trends in women's economic equality. It can help us discern whether there has been a steady increase in improvement, as suggested by the connection between Burnette (2008) and Goldin (1989), or whether the Industrial Revolution marked the beginning of the exclusion of women, a trend not reversed until the 20th century.

Despite its significance, women's experiences during this period have been understudied. Limited availability of reliable data on women's employment has led researchers to exclude them from analysis. Our study aims to address this gap by examining women's labor market participation and occupational choices in the period following the Industrial Revolution, utilizing newly digitized census microdata for England.

The digitization of the Census Enumerators' Books (CEBs) from 1851 to 1911, with the exception of 1871, has opened up an unprecedented opportunity for systematic examination of women's work and its impact on both the market and household economy. These CEBs contain census manuscripts comprising individual-level data on name, sex, age, marital status, "Rank, profession or occupation," place of residence, and various other individual and household characteristics. It's worth noting that the CEBs serve as the primary records used to compile the aggregated Census tables that were accessible before the digitization process. Notably, the reported Census tables underwent extensive revisions before publication, setting them apart from the raw data found in the CEBs.[1]

[1]The accuracy of British census statistics, particularly regarding the recording of women's employment, has been a subject of scrutiny in the literature. Works such as Higgs (1987), highlighted by You (2020b), raised awareness of potential pitfalls in using these statistics. Subsequent studies (Bourke, 1994; Burnette, 2008; Davidoff and Hall, 2018; Hill, 1993; Horrell and Humphries, 1995; Sharpe, 1995; Horrell and Humphries, 1995) have echoed these concerns. Recently, Higgs revisited his earlier claims, offering new insights into the

The CEBs are arguably the most comprehensive source for studying women's employment in the nineteenth century. The volume of data minimizes concerns regarding potential statistical artifacts associated with the 'small sample problem.' Additionally, the national coverage of the CEBs helps circumvent selection issues often encountered in more localized studies. However, this data remains largely unexplored for studying female employment in this period. To the best of our knowledge, we are the first to attempt to construct a reliable series of women's employment using this dataset.

We begin by documenting several empirical facts about women's employment between 1851 and 1911. First, we show that women's labor force participation rate declined during this period. This decline is driven by married women, who experienced a 40% reduction in labor force participation between 1851 and 1911. Conversely, single women saw a small increase in their participation rates over the same period. Second, our analysis indicates that the decline in married women's participation is primarily driven by younger cohorts opting not to enter the labor force, rather than existing cohorts exiting at higher rates – a trend not observed among single women.

We explore potential explanations for these empirical patterns, considering the possibility of an emerging male breadwinner household model. However, our analysis suggests that this hypothesis is unlikely. Similar trends are observed across women of various socioeconomic classes, as indicated by their husbands' occupations, and there is an increasing trend in the employment of children over time. Moreover, changes in household structures do not appear to account for the decline in married women's labor force participation. Thus, we posit that institutional constraints and labor legislation played a more significant role in limiting women's opportunities during this period.

Finally, our analysis of employment sectors reveals that women were predominantly employed in the textile and domestic service industries. Married women were particularly concentrated in textiles, whereas single women were more prevalent in domestic services. Notably, the decline in married women's employment within the textile sector accounts for the majority of the overall decline in labor force participation.

In order to better understand the forces contributing to both the decline in married women's labor force participation and the limited range of occupation choices available to women, we develop a theoretical model. This model uses a simplified version of the framework of Hsieh et al. (2019) (hereafter HHJK), which is a Roy model of occupation choice. The HHJK model introduces group-specific barriers that create misallocation across occupations. For instance, a large wage gap for women may lead them to prefer working at home rather than entering the formal labor market. Similarly, societal norms that prioritize women

reliability of census returns (Higgs and Wilkinson, 2016).

staying at home may discourage women from participating in the labor market compared to men. While wage data necessary to fully characterize the various barriers introduced in the model is unavailable, the available data and model are consistent with the hypothesis that married women are facing increasing wedges or barriers over time.

Related literature. Our paper contributes to two strands of the literature. The first strand examines the role of women in England between 1851 and 1911 using the newly digitized CEBs. Studies such as You (2020a) and You (2020b) have delved into this area. The former, based on the 1881 CEBs, emphasizes the importance of demand-side factors in determining female labor force participation rates during the nineteenth century in England and Wales. The latter, using data from all waves of the CEBs, focuses on the accuracy of married women's employment. In contrast, our paper utilizes data from all waves of the CEBs and employs a theoretical model to analyze the evolution of barriers to women's employment over time.

The second strand of the literature is dedicated to documenting trends in women's historical labor force participation and exploring factors that might explain these changes (Goldin, 1989, 2006; Fernández, 2013; Goldin, 1990; Goldin and Katz, 2002; Attanasio et al., 2008; Vidart, 2024; Greenwood et al., 2005). This body of work typically documents the gradual inclusion of women in the labor force. For example, Goldin (1989) finds that increases in female labor force participation primarily occur through newer generations entering rather than older generations joining at greater rates later in life. Our paper contributes by examining how women's opportunities were impacted differently (in the opposite direction) in a distinct historical era.

The paper proceeds as follows: Section 2 describes the data, Section 3 presents empirical facts about women's employment between 1851 and 1911 in England, Section 4 describes the model and presents estimation results, and Section 5 concludes.

2 Data

The CEBs are the historical British census manuscripts containing individual-level data on sex, age, marital status, "rank, profession or occupation", place of residence and other individual and household level characteristics. While the CEBs were used as an input into the Census tables, the information they provide is not necessarily the same. As described by You (2020b), Census enumerators distributed the household schedules one week before census night. The morning after census night, Census enumerators transferred the information to their CEBs with certain revisions and checks, according to the instructions given to them. The CEBs were then sent to the local registrars and superintendent registrars. At each

stage examinations, checks and simple tabulations were performed (Woollard, 1998). Then the CEBs, household schedules, and various summary forms were sent to the Census Office in London. With a set of instructions given to them, some 60 clerks spent around three months undertaking further checks on the CEBs and performing the tabulations that led to various tables in the published census reports. The only notable exception to the aforementioned procedures occurred in 1911 when the original household schedules rather than the CEBs formed the basis for the tabulation in the published census reports.

The Integrated Census Microdata (I-CeM) Project, based within the Department of History at the University of Essex, created a digitized and standardized data set of all the CEBs between 1851 and 1911. They reformatted the input data, performed a number of consistency checks, coded the nonstandard textual occupational strings with the occupational classification, and added a number of enriched variables relating to household structure (Higgs et al., 2013).

We use the 100 percent sample of the CEBs in England between 1851 and 1911, with the exception of 1871 where information is not available for England. We access the data through IPUMS International Center (2020). We restrict the sample to men and women with ages between 15 and 65, since this is the portion of the population most likely to be active in the labor force. The final sample contains 91,863,887 observations, 52% of which are observations pertaining to women.

A convenient feature of the digitized data is that occupations are mapped into the Historical International Standard Classification of Occupations (HISCO) codes. HISCO is a historically sensitive and internationally valid occupational classification allowing researchers from a variety of countries to communicate with each other and make international comparisons across the nineteenth and twentieth centuries in social, economic and other fields of history. HISCO has a tree-like structure with 9 'major' groups, 76 'minor' groups, 296 'unit' groups and 1675 'micro' groups. The 'leaves' of the tree are formed by the tens of thousands of occupational titles that fall under these micro groups (Leeuwen et al., 2002). We use the 9 'major' groups - agriculture, textiles production, paper production, shoe production, clerical services, domestic services, sales, administrative and managerial, law and arts, and science - and refer to them throughout the paper as sectors.

2.1 Women's occupations in the CEBs

The main challenge we face is in obtaining a consistent series of the labor force participation rate for married women. You (2020b) examines one particular issue pertaining to married women's work: the enumeration of "occupation's wife" in the CEBs, where the occupation

within the quotes refers to the husband's occupation. For instance, if a woman was married to a farmer, the occupational title of the wife would be "farmer's wife." Unlike more conventional occupational titles such as "spinner" or "domestic servant," the term "occupation's wife" is embedded with conceptual ambiguity. The tabulation of "occupation's wife" was also inconsistent over time.

Prior to 1881, Census enumerators were instructed to assume that wives of farmers, shoemakers, butchers, shopkeepers, and innkeepers worked together with their husband in the same trade. From 1881 onward, they were assumed to be unoccupied. However, the term "occupation's wife" was used more generally, even in the later Census waves. Importantly, the instructions explicitly said that this term should not be used if the wife had an independent occupation. You (2020b) shows that almost all occupations had the corresponding enumeration of "occupation's wife" in the CEBs, even if the occupations offered almost no opportunities for married couples to work together[2]. Table 1 reports the share of married women with any occupational title enumerated as "occupation's wife." It illustrates that the coding was inconsistent over time, and that such married women constituted a large share of those with any occupational titles in the earlier Census waves.

You (2020b) also shows evidence suggesting that the enumeration of "occupation's wife" was heavily influenced by an enumerator's idiosyncratic beliefs, possibly as a result of Victorian domestic ideologies. He concludes that the term gives little indication of the true status of married women's labor, whether they work with their husband or are simply out of the labor force. This suggests that using "occupation's wife" to determine whether married women were in the labor force will generate an inaccurate time series of married women's labor force participation rates.

We choose to exclude all individuals whose occupation is "occupation's wife" from the employment totals. We only consider individuals that worked regularly and had independent occupations. This approach likely underestimates married women's labor force participation rates, since it is probable that some of these wives were actually engaged in the same trade with their husbands. In Section 3.3, we perform robustness checks on this exclusion. In particular, we classify a set of occupations where it is plausible that women worked with their husband. In the robustness section we describe in more detail how the occupations are chosen. We then look at how our results would differ if we assumed that 50% (rather than 0%) of wives of husbands working in these areas are active in the labor force, and find that our results remain largely unaffected.

[2]In our sample, 70 out of 76 of 'minor' HISCO codes have the corresponding enumeration of "occupation's wife"

3 Empirical analysis

We start by documenting trends in labor force participation between 1851 and 1911 in England. We observe stable participation rates for men and single women during this period, while participation among married women declined by approximately 40%. We investigate potential explanations for this decline among married women. We end this section by examining the occupational distribution of women during this time period.

3.1 Trends in labor force participation

Figure 1 displays changes in men and women's labor force participation between 1851 and 1911 relative to 1851. The solid lines represent overall participation, with dotted lines indicating single individuals and dashed lines representing married individuals. Table A.1 reports the labor force participation rates in levels.

Figure 1a illustrates that men's participation remained relatively stable, and that there were no large differences between single and married men. Though it fell in 1861 and increased in the subsequent decades, the magnitude of these changes is very small, with the levels of labor force participation fluctuating between 93% and 96% during this period. In contrast, Figure 1b shows a steady decline in women's participation, primarily driven by a nearly 40% decrease among married women by 1911 compared to 1851 levels. However, single women experienced slight increases in participation over the same period.

Women's labor force participation by cohort

Next we investigate reasons behind the decline in the participation of married women in the labor force. First, we investigate whether this decline is driven by older married women exiting the labor market at increasing rates or by younger cohorts of married women choosing not to enter the labor market.

Distinguishing between these two hypotheses is crucial, as they could reflect different barriers to women's work. For instance, if social norms around married women working were changing over time, younger cohorts of married women might be discouraged from entering the labor force at younger ages. Conversely, if societal norms discouraged mothers from working, we would not expect to see married women failing to join the labor force when younger. Instead, we would anticipate women at later points in their life cycle dropping out at increasing rates.

To investigate these two hypotheses, we analyze trends in labor force participation by cohort and age. We create pseudo-panels by tracking synthetic cohorts from 1851 to 1911.

We define five age groups within a cohort's life cycle: very young (age 15-24), young (age 25-34), middle (age 35-44), old (age 45-54), and very old (age 55-64). For example, individuals born between 1827 and 1836 form a synthetic cohort that is very young in 1851, young in 1861, old in 1881, and very old in 1891. There are eleven cohorts over the study period, with observations at four life cycle points for four cohorts (born between 1827-1836, 1837-1846, 1847-1856, and 1857-1866) and fewer life cycle points for the remaining cohorts.

Figures 2a and 2b depict the labor force participation rates of married and single women cohorts over the life cycle, respectively. The observations represent the midpoints of the five age groups, and cohorts are labeled based on their years of birth. Regarding the hypothesis suggesting that younger cohorts are discouraged from entering the labor force, we anticipate a parallel downward shift of the labor force participation rate curve. Conversely, for the hypothesis proposing that women are encouraged to exit the labor force later in life at increasing rates, we expect to observe changes in the slope of the labor force participation rate curve.

Figure 2a shows that younger cohorts of married women consistently exhibit lower labor force participation rates compared to older cohorts at each stage of their life cycle. This suggests a pattern of parallel shifts up or down in the life-cycle labor force participation profiles of different cohorts, supporting the hypothesis that younger cohorts fail to enter the labor force. Additionally, we observe a consistent decline in labor force participation rates for each cohort of married women throughout their married life, indicating that as married women have children, they tend to exit the labor market and do not return.

In contrast, Figure 2b shows that single women display similar life-cycle profiles across cohorts. This indicates that different cohorts of single women enter and exit the labor force at similar rates. Furthermore, the life-cycle labor force participation profiles of single women are concave, with participation rates remaining high until around age 40 and declining thereafter.

Together, Figures 2a and 2b illustrate that labor force dynamics were distinct for married and single women in England between 1851 and 1911.

The evidence presented in Figure 2a is only suggestive. Ideally, we would like to decompose the observed labor force participation rates of different age groups in different years into an additive set of age, year, and cohort effects. Formally, let lfp_{at} is the labor force participation rate of individuals in age group a at time t. Note that people who are in age group a in year t were born in $t-a$: hence $t-a$ is a convenient index for their cohort. We would like to estimate

$$lfp_{at} = \alpha_a + \gamma_{t-a} + \tau_t + \epsilon_{at} \tag{1}$$

where α_a are age effects (for 10-year age bands), γ_{t-a} are cohort effects (for 10-year birth cohorts), and τ_t are year effects (corresponding to the census years).

It is well known that the parameters measuring the age, cohort, and period effects in equation 1 are not separately identified. Therefore, we estimate equation 1 without the year effects, attributing all changes in the dependent variable to age and cohort effects. We employ least squares weighted by the cohort size and cluster the standard errors at the cohort level.

Columns (1) and (2) of Table 2 display the estimated cohort effects. The dependent variables are the labor force participation rates of single and married women, respectively. As expected, in Column (1), we observe no statistically significant differences in the labor force participation rates of single women across different cohorts, relative to the cohort born between 1797 and 1806.

In contrast, when considering married women in Column (2), we find that only younger cohorts exhibit statistically significant differences in labor force participation rates compared to the cohort born between 1797 and 1806. The effects become significant for the cohort born between 1827 and 1836, increasing in magnitude thereafter. Specifically, the labor force participation rate of married women born between 1827 and 1836 is two percentage points lower than the reference cohort, while that of those born between 1887 and 1896 is ten percentage points lower. These effects are meaningful, given that the average labor force participation rate of those born between 1797 and 1806 is 39%.

In order to include year effects into the regression, we can assume that there are no cohort effects for single women, which suggests the single women's labor force participation is of the form:

$$lfp_{at} = \gamma + \alpha_a + \tau_t + \epsilon_{at} \tag{2}$$

That is, we restrict all cohorts of single women to have the same effect on labor force participation. Then, assuming that aggregate economic conditions affected single and married women's labor force participation rates in the same way, we stack data for single and married women and estimate the following regression:

$$lfp_{gat} = \alpha_{ga} + \gamma_{g,t-a} + \tau_t + \epsilon_{gat} \tag{3}$$

where $g \in \{s, m\}$ denotes single or married women. Equation 2 implies that $\gamma_{s,t-a} = 1$ for all $t - a$.

The estimated cohort effects of married women are presented in Column (3) of Table 2. These results are consistent with those of Column (2), where year effects were not included. We observe that younger cohorts of married women exhibit significantly lower labor force participation rates, even after adjusting for age and year effects, when compared to

the cohort born between 1797 and 1806. Furthermore, the negative impact intensifies for younger cohorts of married women. Notably, these effects appear to be even larger than those observed in Column (2).

Next, we investigate reasons behind the empirical patterns described thus far. The fall in married women's labor force participation may be linked to the rise of the male breadwinner family, that is, one in which men supported dependent wives and children (Horrell and Humphries, 1997), as men's earnings increase. Changes in family structure variables could also have been drivers behind the fall in married women's employment. Alternatively, the fall in married women's labor force participation could have occurred due to demand factors. For example, institutional constraints on women's employment, such as exclusionary strategies of trade unions, and labor legislation that limited the opportunities for women. In the remainder of this section, we present empirical patterns that suggest that the fall in married women's labor force participation was not likely due to a rise of the male breadwinner family nor changes in household structure.

Male occupations and wages

We begin our analysis by investigating whether income effects from men's earnings are associated with the decline in married women's labor force participation. Specifically, if husbands' earnings were a key factor driving married women to stay at home, we would expect to observe a larger decline in labor force participation among women married to men in higher-paying occupations. This is because these are the families where men's wages would be sufficient to meet the family budget.

To investigate this hypothesis, we link married couples living in the same household. For the 88% of working-age married women matched to a husband[3], we use information on wages from this time period to classify husbands' occupations as either high or low-paying[4]. In particular, Bowley (2016) and Feinstein (1972) construct an annual wage index beginning in 1855, and Feinstein (1990) provides wage indices starting in 1881, broken down by occupation[5]. We classify women married to men in low-paying occupations as belonging to the lower class, and those married to men in high-paying occupations as belonging to the middle class.

[3]That is, 12% of married women cannot be matched to a husband because they do not live in the same household.

[4]Low-paying occupations include domestic service, agriculture, textile production, mining, shoemaking, cabinetmaking, stone cutting, blacksmithing, glassmaking, and pottery. High-paying occupations include science, law, arts, administration, management, clerical work, sales, machine or electrical work, broadcasting, sound production, cinema work, plumbing, welding, sheet metal work, and jewelry or precious metal work.

[5]Occupation groups available include agriculture, mining, construction, manufacturing, transportation, distribution, domestic service, government, and general labor.

Figure 3a displays married women's labor force participation between 1851 and 1911, using 1851 as the base year. Solid lines represent participation of women in the lower class, while dotted lines indicate women in the middle class. The results do not support the hypothesis that income effects from men's earnings are associated with the decline in married women's labor force participation, as women in both the low and middle classes experience the same decline in labor force participation.

Furthermore, Figures 3b and 3c depict the labor force participation rates of low and middle class married women cohorts over the life cycle, respectively. The observations represent the midpoints of the five age groups, and cohorts are labeled based on their years of birth. We observe a similar pattern for both groups: younger cohorts of married women are failing to join the labor force.

A shift towards a male breadwinner family would also likely result in a reduction in the employment of children, particularly those whose mothers do not work. We investigate whether the data is consistent with this by examining trends in the labor force participation of working-age children of married women, specifically those 15 years or older, who live with their mothers. Figure 4 presents the results, which do not support the hypothesis. Children of married women who are working do not experience any changes in labor force participation between 1951 and 1911. However, children of married women who are not working experience a rise in employment, which is more pronounced for daughters. This suggests that the male breadwinner hypothesis is unlikely to be the cause of the decline in married women's labor force participation. Instead, Figure 4 suggests that children might have had to work increasingly to compensate for their mothers' lost income.

Family structure

The decline in married women's labor force participation could be linked to changes in family structure over our study period. For instance, the trend could reflect a selection effect. Specifically, if women delayed marriage to work, then observing younger cohorts of married women failing to join the labor force at earlier stages in their life cycle could be attributed to the selection of who gets married at those younger ages. However, Figure 5a shows that the share of women who are married at each point in their life cycle is similar across cohorts. This suggests that changes in the composition of married women are unlikely to be responsible for the decline in married women's labor force participation.

Alternatively, changes in fertility patterns could drive our results. For instance, younger cohorts of married women could be failing to join the labor force if they began having children earlier in their life cycle. This would be reflected by a larger share of married women having children earlier in their life. Figure 5b shows that the share of married women living with

at least one child at different points in the life cycle is similar across cohorts. Moreover, the share of married women with children rose by only 3 percentage points, from 19% to 22%, between 1851 and 1911. This suggests that changes in fertility patterns are not the primary explanation for the labor force participation results.

While we don't observe changes in the family household structure, the decline in married women's labor force participation could be linked to changes in social norms and perceptions about working mothers. Figure 6a plots married women's labor force participation between 1851 and 1911, using 1851 as the base year, separately by whether she has at least one child (dashed line) or not (solid line). We observe strikingly similar trends in labor force participation for both groups, indicating that changes in views about working mothers are unlikely to be the leading explanation.

Finally, changes in the age profile of children could be responsible for the observed patterns in married women's labor force participation. Figure 6b plots married women's labor force participation between 1851 and 1911, using 1851 as the base year, separately by the age profile of her children. We define young children as those under 5 years old. Mothers with young and old children living in the household, and only young or only old children, all experience large declines in labor force participation during this period. Therefore, even if fewer mothers have older children in the household to help with childcare of younger siblings, this explanation would not be able to explain the decline in married women's labor force participation between 1851 and 1911.

We conclude that neither a rise in the male breadwinner model nor changes in household structure are key drivers of the decline in married women's labor force participation between 1851 and 1911. Instead, the empirical analysis points to institutional constraints on women's employment, such as exclusionary strategies of trade unions, and labor legislation that limited opportunities for women, as potential explanations for the decline in married women's work opportunities.

3.2 Trends in occupation distribution

In this section, we examine the occupational distributions of men and women between 1851 and 1911. Figures 7a, 7b, and 7c display the occupation distribution of married women, single women, and men, respectively, in 1851, 1881, and 1911. A full description of the occupations within each occupation group can be found in Appendix D.

Figures 7a and 7b show that women primarily worked in textiles and domestic service. The concentration of women in these two main industries highlights the limited scope of opportunities for women's work during this period. Additionally, we observe that married

women were more concentrated in textiles, while single women were more concentrated in domestic services. One explanation for this difference is that domestic service workers were often required to live with their employer, and single women were freer to live away from their families. Specifically, 71% of single women employed as domestic servants lived with non-relatives, while only 23% of single women employed in other sectors lived with non-relatives. This contrasts with the fact that only 16% of married women employed as domestic servants lived with non-relatives.

Figure 7a shows that between 1851 and 1911, the share of married women working in textiles decreased. Meanwhile, married women became more likely to be employed in domestic services. Figure 7b indicates that textiles also employed relatively fewer single women. However, unlike married women, single women were also less likely to be employed in domestic services by 1911. Single women appear to have gained opportunities to work in sales and clerical positions between 1851 and 1911.

The observed decrease in the share of married and single women working in textiles could be due to a decline in the industry over this period or increasingly stringent laws that restricted the number of hours women and children were allowed to work in factories. For instance, in the 1840s, a series of Factory Acts were put in place to protect both children and women from harsh working conditions. This may have pushed women out of these industries. Moreover, women were not allowed to enter the skilled male unions during this period (Holloway, 2007), which might have further restricted work opportunities for women.

Figure 7c illustrates that men's occupation distribution was less concentrated than that of women. We observe a decrease in the share of men employed in textiles, suggesting a decline in the textile industry during this period. Additionally, the proportion of men working in agriculture fell by over 50% between 1851 and 1911. Men increasingly shifted towards occupations such as sales, clerical tasks, and paper, which accounted for a relatively small share of female employment.

One hypothesis for the decline in married women's labor force participation over this period suggests that if it was easier for them to enter the formal labor force by working in the same occupation as their husband, then a decrease in opportunities to work in the same trade as their husband could explain this trend. We investigate this hypothesis in Table 3, which reports the share of women employed in the same occupation as their husband, categorized by her occupation. We observe that in 1851, 46.7% of married women employed in textiles had husbands working in the same trade. Conversely, only 8.3% of married women employed in domestic services had husbands working in this sector in the same year.

Furthermore, Column (7) of Table 3 indicates that across all occupations, except for Law/Arts and Domestic Services, the share of women working in the same occupation as

their husband decreased between 1851 and 1911. Therefore, a decrease in opportunities for married women to work with their husband may potentially help explain the decline in married women's labor force participation observed in Figure 1b.

3.3 Robustness Checks

We conduct a series of robustness checks. First, we examine whether our results are sensitive to the decision of considering married women with independent occupations only. This excluded women that worked with their husbands, who were inconsistently attributed the term "occupation's wife" in the 1851, 1861 and 1911 Censuses, and were coded as out of the labor force in the other Census waves. The exclusion of such women likely underestimates the number of married women that were in the labor force. Therefore, we consider how the trend in labor force participation and occupation distribution of married women would change if we count a share of them as working.

We consider the ten most common corresponding occupations associated with the term "occupation's wife" using the 'minor' HISCO codes in 1851. These are agricultural and animal husbandry workers, farmers, carpenters, wholesale and retail trade workers, spinners, weavers, knitters, dyers, miners, shoemakers and leather good makers, blacksmiths, toolmakers and machine-tool operators, transport equipment operators, tailors, dressmakers, sewers, upholsterers. Given that You (2020b) documents that the enumeration of "occupation's wife" was heavily influenced by enumerators' idiosyncrasy, we consider the sample of potential workers as all married women without independent occupations that are married to individuals with one of these occupations.[6] We then assume that 50% of all these women are in the labor force, and attribute to them the same occupation as their husband.

Figure A.1a illustrates the changes in labor force participation trends as a result. Although the magnitude of the decrease is now smaller, married women's labor force participation rate continued to fall relative to its 1851 level. By 1911, married women's labor force participation rate had fallen by 25% compared to 1851. Figure A.1b displays the resulting occupation structure. Married women remained heavily concentrated in textiles, with domestic services representing a lower share of employment. This change in pattern was driven by the increase in married women working in sales, retail trade (shoes and paper), and agriculture alongside their husbands.

Although there are a few differences in the occupation structure of married women when we include women who were more likely to have genuine labor force attachment, the qual-

[6] We are only able to link couples that live in the same household. We match 88.3% of married women to their husband.

itative results remain unchanged. Married women's work opportunities remained highly concentrated in textiles, and their labor force participation rate declined between 1851 and 1911.

The second robustness check relates to the regression exercise, where we examine the trend in married women's labor force participation by cohort. In particular, we estimate equations 1 and 3 at the individual level. The results are displayed in Table A.2. We can see that the results are qualitatively similar to those displayed in Table 2.

4 Model

Two key facts emerge from our empirical analysis. The first, that younger generations of married women fail to participate in the labor force. The second, that for both single and married women, relatively few occupations were available to them for formal, and independent work. This section builds a model to formalize the link between these phenomena.

There is a modern macro-labor literature looking at why individuals are misallocated across occupations. In particular, Hsieh et al. (2019) develops a model that views the source of the misallocation as a set of wedges faced by different minority groups. For instance, minority groups may face different societal pressures, lower wages, or larger barriers to human capital accumulation relative to their majority counterparts. The relative wedges in different occupations determine how likely it is that an individual will choose a given occupation, or choose to be in the labor force in general.

We use a simplified version of the model in Hsieh et al. (2019) to look at the relative barriers faced by single and married women relative to men. Our model will include two wedges capturing i) the relative preferences of society, and ii) the relative wages paid to the various groups. The size of these relative barriers, and how they have changed over time, can help us think about the empirical patterns found above. What do the labor force participation trends found above say about the societal preferences of women's work in Victorian England? What do the trends in choices over different occupations tell us about the widening (or narrowing) of women's pool of available occupations?

4.1 Model set-up

As in HHJK, the model will have a continuum of workers. Here, workers will belong to one of four groups $G = \{sw, mw, sm, mm\}$, which correspond to single women, married women, single men and married men, respectively. A given individual's group is denoted g. There

are I occupations for a worker to choose from. Working from home (or being out of the labor force) is considered one of these I occupations.

As in a traditional Roy (1951) model, workers choose the occupation that maximizes their utility. Workers are born in period c and live for three periods. Individuals work in their chosen occupation for all three periods. The lifetime utility associated with choosing occupation $i \in I$ is:

$$\log U_i = \sum_{t=c}^{c+2} \log C_i(c,t) + \log z_{ig}(c) \tag{4}$$

Note that indices for individuals are suppressed for ease of notation.

There are two forces which will affect the utility of a given occupation. The first force is the utility gained from 'societal approval'. This is the first wedge faced by different groups, and is denoted z_{ig}. It generates a utility benefit for everyone belonging to group g who works in occupation i. For instance, if Victorian Era society believes married women should work in the home rather than in the formal labor market, this will be large for $i = home$, $g = mw$, but small for $i \neq home$, $g = mw$. An individual makes a choice given the societal beliefs at the time they are born, $z_{ig}(c)$.

The second force is how much an individual is able to consume in that occupation, $C(c,t)$. In this model there is no mechanism for savings, and agents simply consume their labor income. labor earnings reflect both take-home wages and the efficiency units an individual provides to the market:

$$C_i(c,t) = \underbrace{[1 - \tau_{ig}^w(t)]w_i(t)}_{\text{Take-home wages}} \underbrace{\bar{h}_{ig}\gamma(t-c)\epsilon_i}_{\text{Efficiency units}} \tag{5}$$

Here, the take-home wage has two components. The common factor $w_i(t)$ represents the wage in occupation i at time t. $1 - \tau_{ig}^w(t)$ represents the wage gap in occupation i for group g at time t, and is the second wedge faced by different groups. The efficiency units portion also has multiple parts. $\bar{h}_{ig}$ represents average talent for group g in occupation i; this can be thought of as natural proclivity of some groups to certain occupations. For instance, for occupations with manual labor requiring a lot of physical strength, this parameter may be higher for men than for women. $\gamma(t-c)$ captures how skill varies over the life-cycle. Lastly, ϵ_i is the individual's idiosyncratic talent in the occupation.

We assume that the vector of idiosyncratic talent is drawn from a multivariate Fréchet distribution. That is:

$$F(\epsilon_1, ..., \epsilon_I) = \exp\left\{ -\sum_{i=1}^{I} \epsilon_i^{-\theta} \right\}$$

Note that this is equivalent to talent in each occupation being drawn iid from a univariate Fréchet. The parameter θ controls the dispersion of talent, where a higher θ is associated with lower dispersion. This assumption is necessary for all the results derived later; these would not hold under alternative distributions.

In order to evaluate their lifetime utility in an occupation, agents need to be able to forecast how consumption (i.e. wages) will move over their lifetime. We make the simplifying assumption that agents believe that wages (and wage barriers) for each occupation will remain at the level they initially observe. Thus, expected utility for each occupation can be written as:

$$U_i = \left(\bar{z}_{ig}(c)[1-\tau_{ig}^w(c)]w_i(c)\bar{\gamma}\bar{h}_{ig}\epsilon_i\right)^3$$

Where $\bar{\gamma} = (\gamma(0)\gamma(1)\gamma(2))^{\frac{1}{3}}$ and $\bar{z}_{ig}(c) = z_{ig}(c)^{\frac{1}{3}}$.

For the baseline model, we have no differences in consumption or utility for married versus single individuals, i.e. married people will not pool their resources. This is done for tractability and we hope in later versions to relax this assumption. Thus, differences in married people vs. single people working will be a function only of wedges, not from lack of need to work due to spousal income.

Occupation choice

An individual will choose the occupation with the highest expected utility. Note that the larger $z_{ig}(c)$ the higher the utility of that occupation, and the more likely it gets chosen. Conversely, the larger τ_{ig}^w the lower the utility of that occupation, and the less likely it gets chosen. Thus the share of group g going to occupation i will be increasing in how much society approves of it and decreasing in its wage gap. This is formalized in Proposition 1.

Proposition 1 (Occupation choice)**.** *The share of individuals in group g in cohort c who choose occupation i is described by:*

$$p_{ig}(c) = \frac{\tilde{w}_{ig}(c)^\theta}{\sum_{i=1}^I \tilde{w}_{ig}(c)^\theta}$$

Where $\tilde{w}_{ig}(c) = \bar{\gamma}\bar{z}_{ig}(c)[1-\tau_{ig}^w(c)]w_i(c)\bar{h}_{ig}$
The proof is in Appendix A.

How would we think of these wedges in terms of our empirical findings? The reason for seeing women concentrated primarily in a few occupations could be due to high societal approval of women working in those occupations (or low societal approval for women working in other occupations). It could also be due to better wages for women in those occupations.

Similarly, the decrease in married women's labor force participation could be due to decrease in societal approval in common women's occupations, or to an increase in societal approval of married women working at home. It could also be due to growing labor wedges where married women commonly work. In the following sections, we derive a set of moments that can be observed in the data which will help distinguish between the wage and societal approval effects.

Average wages

One moment that is (potentially) observed in the data is average wages across groups and occupations. Proposition 2 derives average wages in period t for individuals of cohort c in group g who chose occupation i. This will be the take-home wage for individuals in group g in occupation i at time t times the average efficiency units.

Proposition 2 (Average wage)**.** *Average earnings at time t for individuals in group g who choose occupation i are as follows:*

$$\begin{aligned}
\overline{wage}_{ig}(c,t) &= [1-\tau^w_{ig}(t)]w_i(t)\mathbb{E}\left[\gamma(t-c)\bar{h}_{ig}\epsilon_i | chooses\ i\right] \\
&= [1-\tau^w_{ig}(t)]w_i(t)\gamma(t-c)\bar{h}_{ig}\left(\frac{1}{p_{ig}(c)}\right)^{\frac{1}{\theta}}\Gamma\left(\frac{\theta-1}{\theta}\right) \\
&= \frac{1-\tau^w_{ig}(t)}{1-\tau^w_{ig}(c)}\frac{w_i(t)}{w_i(c)}\frac{\gamma(t-c)}{\tilde{\gamma}}\tilde{z}_{ig}(c)^{-1}\left(\sum_{i=1}^{I}\tilde{w}_{ig}(c)^{\theta}\right)^{\frac{1}{\theta}}\Gamma\left(\frac{\theta-1}{\theta}\right)
\end{aligned}$$

The proof is in Appendix A.

There are two ways average earnings of group g in occupation i at time t can increase. One is the direct effect of movements in the take-home wage, either $\tau^w_{ig}(t)$ decreases or $w_i(t)$ increases. Another is if average efficiency of the group increases.

The average efficiency is inversely related to the proportion of people in that occupation. If there are very few individuals of group g in a given occupation, those who chose that occupation must have high idiosyncratic talent, ϵ_i. As more people join, they will on average have lower talent. As discussed above, the likelihood of joining a given occupation is related to the size of the two wedges at the time the cohort is born. That is, if τ^w_{ig} is high at time c, less people in g will join occupation i and average efficiency will be higher. Similarly, if $\bar{z}_{ig}$ is high at time c, more people in g will join occupation i and average efficiency will be lower. This is why average earnings are inversely related to these wedges.

Identification of wedges with ideal data

Propositions 3 and 4 show how we can identify the τ versus z wedges using data on proportions of groups in different occupations, and average wage data of groups in different occupations.

Proposition 3 looks at the propensity for group g to work in the home sector (i.e. out of the labor force), relative to the propensity for single men to do so. We normalize the home sector to have no gaps for all groups.

Proposition 3 (Relative LFP)**.** *If there are no gaps in the home sector, that is,* $\bar{z}_{home} = 1$*,* $\tau^w_{home} = 0$*, and* $\bar{h}_{home} = 1$*, then relative home sector employment is as follows:*

$$\begin{aligned}\frac{1 - LFP_g(c)}{1 - LFP_{sm}(c)} &= \frac{\sum_{i=1}^{I} \tilde{w}_{sm}(c)^\theta}{\sum_{i=1}^{I} \tilde{w}_{i,g}(c)^\theta} \\ &= \left(\frac{\overline{wage}_{i,sm}(c,c)}{\overline{wage}_{ig}(c,c)}\right)^\theta \left(\frac{\bar{z}_{i,sm}}{\bar{z}_{ig}}\right)^\theta \quad \forall i\end{aligned}$$

The proof is in Appendix A.

Proposition 3 relates relative labor force non-participation to the wage gap and societal preferences gap. Since this relationship holds for every occupation i, average wages across occupations are fully explained by societal preferences. In particular, if average wages are relatively low for a group in an occupation, it must be that societal preferences are relatively high. If both labor force participation and average wages are observed, we can identify relative societal preferences, $\bar{z}$.

Proposition 4 calculates the relative proportion of group g in occupation i relative to the number of single men in that occupation.

Proposition 4 (Relative propensity)**.**

$$\begin{aligned}\frac{p_{ig}}{p_{i,sm}} &= \frac{\tilde{w}_{ig}(c)^\theta}{\sum_{i=1}^{I} \tilde{w}_{ig}(c)^\theta} \frac{\sum_{i=1}^{I} \tilde{w}_{ism}(c)^\theta}{\tilde{w}_{i,sm}(c)^\theta} \\ &= \left(\frac{[1 - \tau^w_{ig}(c)]\bar{h}_{ig}}{[1 - \tau^w_{i,sm}(c)]\bar{h}_{i,sm}}\right)^\theta \left(\frac{\overline{wage}_{ig}(c,c)}{\overline{wage}_{i,sm}(c,c)}\right)^{-\theta}\end{aligned}$$

The proof is in Appendix A.

This proposition helps identify changes in relative wage gap wedge. Assuming that relative natural propensities, $\bar{h}$, do not change over time, the change in the wage wedge can be identified as the change in relative proportions that cannot be explained by relative average wages. As discussed above, Proposition 3 says that average wages will be inversely

related to societal preferences. The wage gap wedge is then the 'residual' of what share of people choosing an occupation cannot be explained by societal preferences.

4.2 Model estimation

A serious limitation of our work is the lack of wage data for the period we are interested in. Ideally, we would want average wages for workers in each of the groups (married/single men/women) and each of the occupations. While we have found some information on wages, none of the references break down their wage index on the basis of sex (Feinstein, 1972, 1990; Bowley, 2016).

With these limitations in mind, it is worth asking what insights we can glean from this model without this wage data. If we assume that the 'benchmark' group, single men, do not face wedges, then the change in relative occupation shares over time describes how the overall 'wedge' has changed over time (wages *and* societal preferences). Essentially we say that the occupation distribution of single men is the 'true' occupation distribution with no misallocation; the closer the other groups are to this the less misallocation there is. Average wage data helps identify whether this overall misallocation stems from wage gaps or from societal preferences.

Figure A.1b plots the variance of $\frac{p_{ig}}{p_{i,sm}}$ over time, relative to 1851. According to Propositions 3 and 4 this variance is related to the variance of the wedges over time. Another way to think of it is a summary measure of how similar the occupation distributions of group g and sm are. As suspected, the occupation distribution of married women is deviating further from, rather than converging to, the occupation distribution of single men. Through the lens of the model, this is an increase in the barriers, or an increase in the misallocation of female labor. Similar trends in misallocation do not appear for single women and married men.

While we cannot quantitatively speak to the source of the wedge, we can give qualitative evidence based on views of women and women's work during this time period. The period we study falls squarely into the Victorian Era in England, which is typically associated with strict views on the roles of women. Women in general, and married women in particular, were thought of in a purely secondary role to their husband. Indeed, their occupation was seen as to care for their spouse and their children. Terms such as 'angel in the house' and the 'household general' are associated with the ideal Victorian housewife: quiet, demure, hard-working, organized. This role for women was made concrete by their lack of legal rights, including for part of this period an inability to hold independent wealth or property.

While this Victorian ideal may seem to exist only for the middle and upper classes, we found evidence that this view was more broadly held. For instance, in the 1840s a series of

Factory acts were put in place to protect both children and women from working conditions. The acts included measures such as limiting working hours, and which days of the week women and children could be asked to work. In 1842, the Mining and Colleries Act specified that no females or children under the age of 10 could have underground jobs in mining. While surely the purpose of these laws was to reduce the devastating affects of child labor, it is telling that these laws also protected women but not men. In Victorian England women, like children, should not be forced to work.

5 Conclusion

The limited availability of reliable data on women's employment has often resulted in the exclusion of women from historical economic studies. The goal of this paper was to utilize newly digitized census microdata for England to examine women's labor market participation between 1851 and 1911. Mokyr (2010) writes that "a considerable literature has maintained that even if the economic position in the pre-industrial era is easily over-idealized as a 'golden age' that never was, the Industrial Revolution was responsible for the marginalization of female labor and the gradual relegation of women to the position of homemaker." Our findings demonstrate that the marginalization of women continued even after the end of the Industrial Revolution.

We documented two facts about women in the labor force. First, women's labor force participation rate slightly decreased between 1851 and 1911. The overall decrease in female labor force participation is entirely driven by younger cohorts of married women failing to join the labor force. This contrasts with the findings of studies on female labor force participation in the 20th century, which show the exact opposite effect: younger cohorts of women progressively join the labor force. Second, we show that women primarily worked in textiles and domestic services. Married women were relatively more concentrated in textiles, while single women were relatively more concentrated in domestic services. The share of single and married women working in textiles declined over time.

Even during the heyday of women's employment opportunities, there was no broad-based inclusion of women. Women were only able to work in a few select occupations, such as domestic services and textile factories, or not at all. Indeed, the occupation distribution of married women is moving further away from that of single women, indicating increasing misallocation of married women's labor due to the growing existence of wedges.

Tables

Table 1: If "occupation's wife" were an occupation

Year	1851	1861	1881	1891	1901	1911
Share	0.42	0.41	0.46	0.03	0.04	0.11

Notes: The table reports the share of married women with any occupational title enumerated as "occupation's wife" in the CEBs. Occupation in the term "occupation's wife" refers to the husband's occupation. For example, if a woman was married to a farmer, the occupational title of the wife would be "farmer's wife."

Table 2: Age-cohort regressions

Cohort	(1) Single	(2) Married	(3) Married
1787-96	-0.0180	0.00323	0.0129**
	(0.0138)	(0.00930)	(0.00466)
1807-16	-0.00398	-0.0119	-0.0150***
	(0.0130)	(0.00824)	(0.00385)
1817-26	-0.00916	-0.0145	-0.0277***
	(0.0153)	(0.00937)	(0.00763)
1827-36	-0.0391*	-0.0234***	-0.0433***
	(0.0200)	(0.00590)	(0.00976)
1837-46	-0.0291	-0.0477***	-0.0665***
	(0.0199)	(0.00610)	(0.00994)
1847-56	-0.0127	-0.0645***	-0.0859***
	(0.0122)	(0.00667)	(0.0120)
1857-66	-0.00282	-0.0795***	-0.0999***
	(0.0190)	(0.00770)	(0.0146)
1867-76	-0.0224	-0.0963***	-0.120***
	(0.0207)	(0.0107)	(0.0178)
1877-86	-0.0172	-0.102***	-0.140***
	(0.0223)	(0.0122)	(0.0220)
1887-96	0.00521	-0.109***	-0.157***
	(0.0255)	(0.0184)	(0.0274)
Year Effect	No	No	Yes
Observations	30	30	60

Notes: The dependent variable is the labor force participation of individuals in cohort c at time t. Column (1) displays cohort-effects of single women and Column (2) and (3) considers those of married women. Standard errors are reported in parentheses and are clustered at the cohort level. The excluded cohort is 1797-1806. All regressions include a full set of age effects (10-year age bands). Regressions are weighted by the cohort size. ***p<0.01, **p<0.05, *p<0.1.

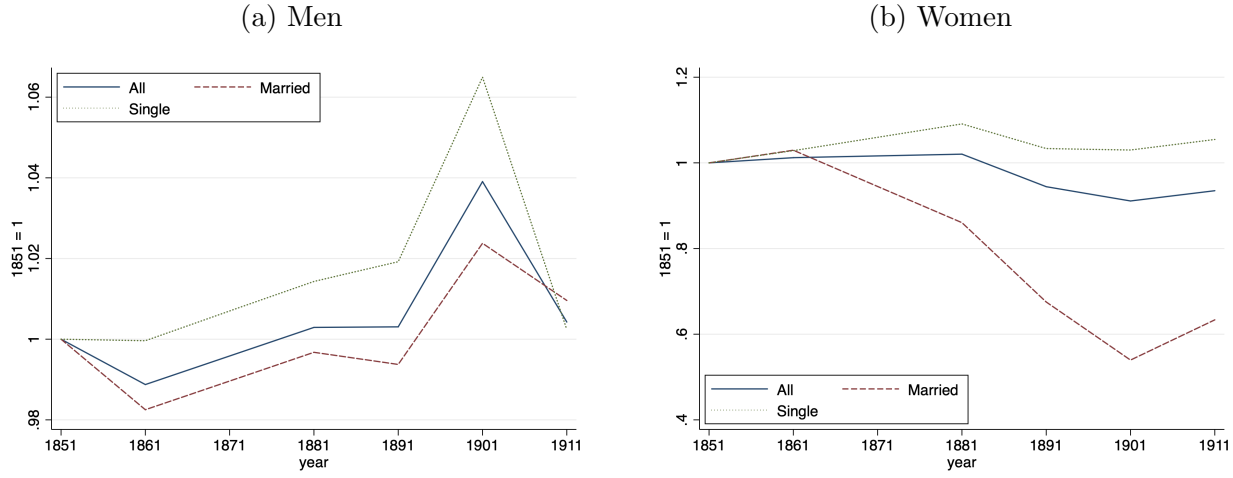
Table 3: Share of women working in the same occupation as her husband, by her occupation

Occupation	(1) 1851	(2) 1861	(3) 1881	(4) 1891	(5) 1901	(6) 1911	(7) Δ
Science	12.8	12.7	9.2	7.8	6.6	5.6	-7.2
Law/Arts	34.3	40.7	47.0	54.2	52.5	44.3	10.0
Administrative/Managerial	35.0	34.5	40.7	29.0	23.4	13.0	-22.0
Clerical	39.2	42.2	40.4	29.8	25.2	23.1	-16.1
Sales	37.9	37.0	36.9	40.2	29.8	31.5	-6.4
Domestic Service	8.3	7.8	8.7	13.6	14.7	15.0	6.7
Agricultural	88.8	86.1	80.0	76.9	63.6	67.7	-21.1
Production (Textiles)	46.7	43.2	38.4	39.3	39.5	38.9	-7.8
Production (Shoes)	64.8	65.6	56.5	53.6	54.2	38.3	- 26.5
Production (Paper)	41.5	42.6	45.5	47.6	45.8	29.5	-12

Notes: The table reports the share of women employed in the same occupation as their husband, categorized by her occupation. That is, among women employed in each occupation, what is the share that have a husband working in the same occupation. Column (7) computes the difference between Columns (6) and (1).

Figures

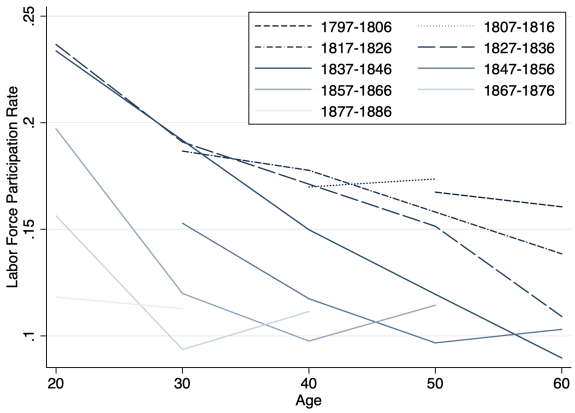
Figure 1: Labor force participation (1851-1911)



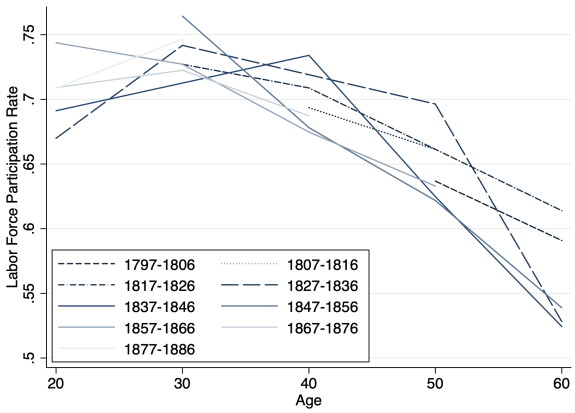
Notes: The figure shows the labor force participation of men (Figure 1a) and women (Figure 1b), calculated using the CEBs. Labor force participation rate is normalized to be equal to 1 in 1851.

Figure 2: Women's labor force participation, by cohort

(a) Married



(b) Single



Notes: The figure shows labor force participation of married (Figure 2a) and single (Figure 2b) women, by cohort. The observations represent the midpoints of the five age groups, and cohorts are labeled based on their years of birth.

References

Orazio Attanasio, Hamish Low, and Virginia Sánchez-Marcos. Explaining Changes in Female Labor Supply in a Life-Cycle Model. *American Economic Review*, 98(4):1517–1552, 2008.

Joanna Bourke. Housewifery in working-class england 1860-1914. *Past & Present*, (143): 167–197, 1994.

Arthur Lyon Bowley. *Wages and Income in the United Kingdom since 1860*. Cambridge University Press, 2016.

Joyce Burnette. *Gender, work and wages in industrial revolution Britain*. Cambridge University Press, 2008.

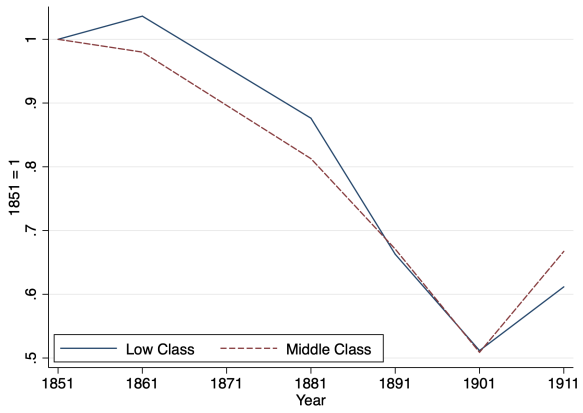
Minnesota Population Center. Integrated Public Use Microdata Series, International: Version 7.3,, 2020. URL https://international.ipums.org. Version Number: 7.3 type: dataset.

Leonore Davidoff and Catherine Hall. *Family fortunes: Men and women of the English middle class 1780–1850*. Routledge, 2018.

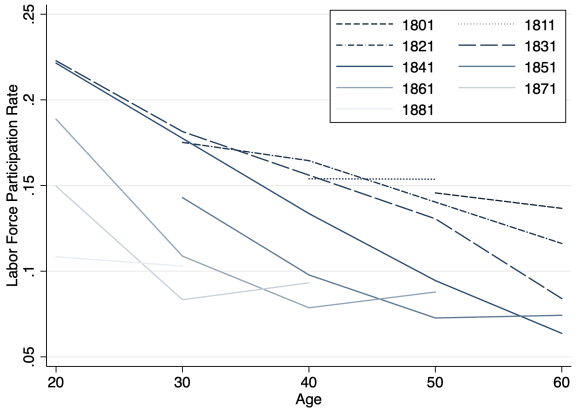
Marie-Louise Décamps. *Essays in Development Economics*. PhD thesis, ProQuest Dissertations & Theses, 2024. ProQuest Dissertations & Theses Global. Database copyright ProQuest LLC; ProQuest does not claim copyright in the individual underlying works.

Figure 3: Married women's labor force participation, by husband occupation

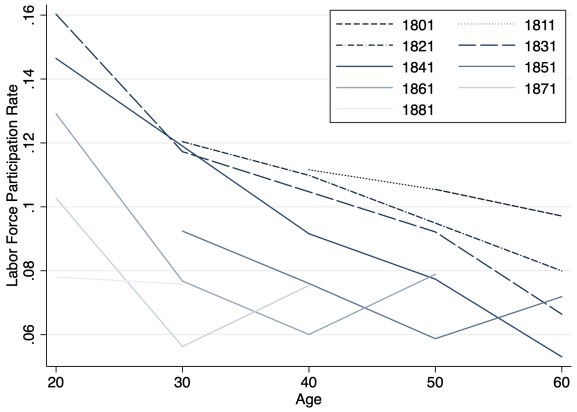
(a) Overall



(b) Low-class, by cohort



(c) Middle-class, by cohort



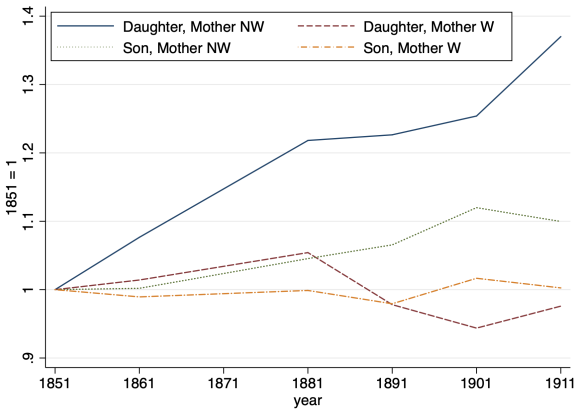
Notes: Figure 3a shows the labor force participation of married women, by low and middle class. Labor force participation rate is normalized to be equal to 1 in 1851. Figure 3b and Figure 3c show labor force participation of married women in low and middle-class by cohort, respectively. The observations represent the midpoints of the five age groups, and cohorts are labeled based on their years of birth. We classify women married to men in low-paying occupations as belonging to the lower class, and those married to men in high-paying occupations as belonging to the middle class.

Charles Feinstein. New estimates of average earnings in the united kingdom, 1880-1913. *Economic History Review*, pages 595–632, 1990.

Charles Hilliard Feinstein. *National income, expenditure and output of the United Kingdom, 1855-1965*, volume 6. Cambridge University Press Cambridge, 1972.

Raquel Fernández. Cultural change as learning: The evolution of female labor force participation over a century. *American Economic Review*, 103(1):472–500, 2013.

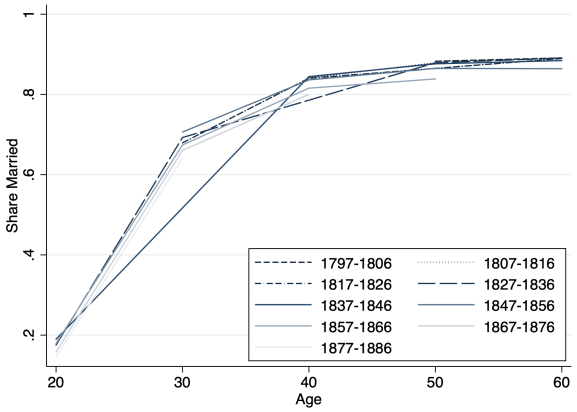
Figure 4: Labor force participation of children of married women, by mother's LFP



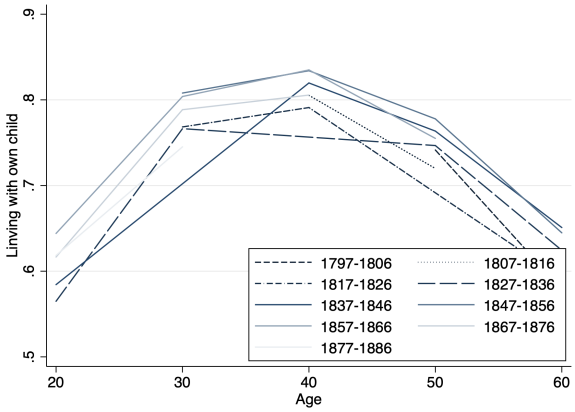
Notes: The figure displays the labor force participation of working-age children of married women, specifically those 15 years or older, who live with their mothers. Mother NW indicates that the mother is not working, while Mother W indicates that the mother is working. Labor force participation rate is normalized to be equal to 1 in 1851.

Figure 5: Characteristics of women, by cohort

(a) Share married



(b) Married and living with own child



Notes: Figure 5a shows the share of women that are married at each age group, by cohort. Figure 5b shows the share of married women that have at least one child at each age group, by cohort. The observations represent the midpoints of the five age groups, and cohorts are labeled based on their years of birth.

Claudia Goldin. Life-cycle labor-force participation of married women: Historical evidence and implications. *Journal of Labor Economics*, 7(1):20–47, 1989.

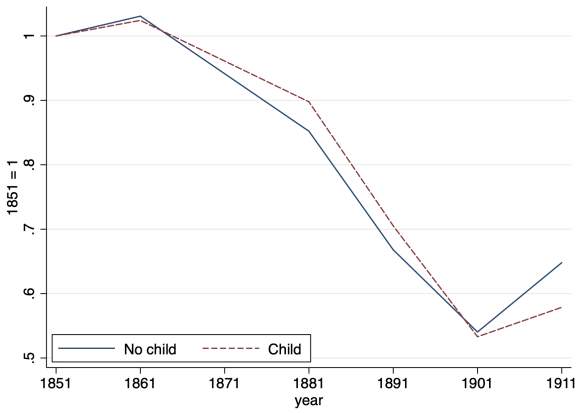
Claudia Goldin. *Understanding the Gender Gap: An Economic History of American Women*. National Bureau of Economic Research, 1990.

Claudia Goldin. The quiet revolution that transformed women's employment, education, and family. *American economic review*, 96(2):1–21, 2006.

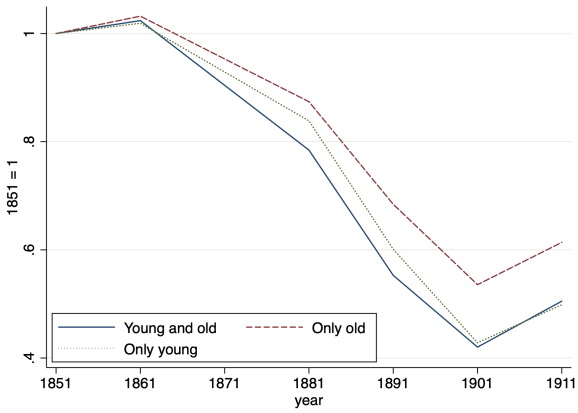
Claudia Goldin and Lawrence F Katz. The Power of the Pill: Oral Contraceptives and

Figure 6: Married women's labor force participation (1851-1911)

(a) By whether women has a child



(b) By age of children



Notes: Figure 6a shows the labor force participation of married women, by whether she has at least one child. Figure 6b shows the labor force participation of married women with at least one child, by the age of her children. Young indicates children under 5 years old. Old indicates children 5 years old or older. Labor force participation rate is normalized to be equal to 1 in 1851.

Women's Career and Marriage Decisions. *Journal of Political Economy*, 110(4):730–770, 2002.

Jeremy Greenwood, Ananth Seshadri, and Mehmet Yorukoglu. Engines of Liberation. *The Review of Economic Studies*, 72(1):109–133, 2005.

Edward Higgs. Women, occupations and work in the nineteenth century censuses. In *History Workshop Journal*, volume 23, pages 59–80. Oxford University Press, 1987.

Edward Higgs and Amanda Wilkinson. Women, occupations and work in the victorian censuses revisited. In *History Workshop Journal*, volume 81, pages 17–38. Oxford University Press, 2016.

Edward Higgs, Christine Jones, Kevin Schürer, and Amanda Wilkinson. Integrated census microdata (i-cem) guide, 2013. https://www.essex.ac.uk/research-projects/integrated-census-microdataf.

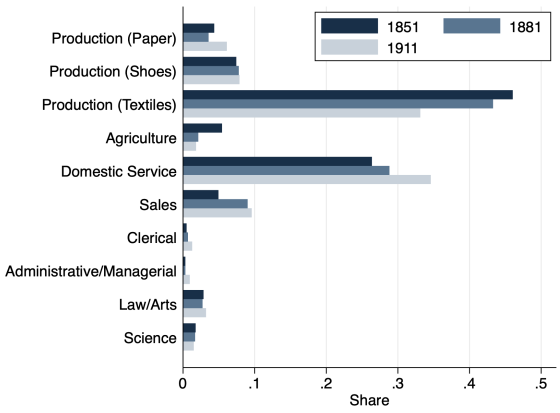
Bridget Hill. Women, work and the census: a problem for historians of women. In *History Workshop Journal*, volume 35, pages 78–94. Oxford University Press, 1993.

Gerry Holloway. *Women and Work in Britain since 1840*. Routledge, 2007.

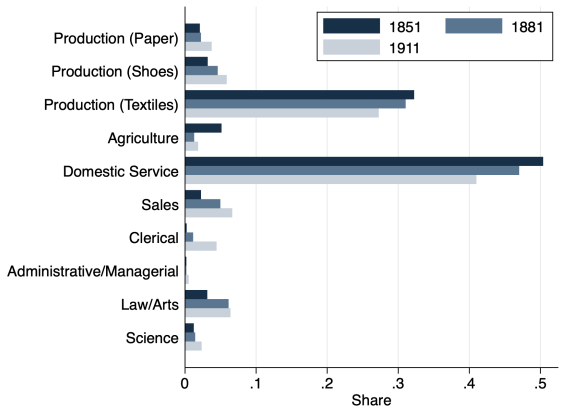
Sara Horrell and Jane Humphries. Women's labour force participation and the transition to the male-breadwinner family, 1790-1865. *Economic History Review*, pages 89–117, 1995.

Figure 7: Occupation Distribution

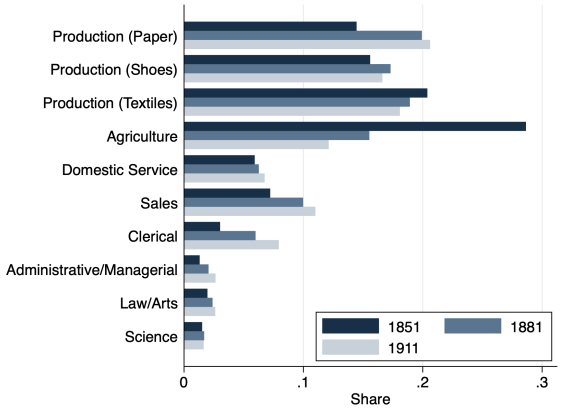
(a) Married women



(b) Single women



(c) Men



Notes: Figures 7a, 7b, and 7c display the occupation distribution of married women, single women, and men, respectively, in 1851, 1881, and 1911.

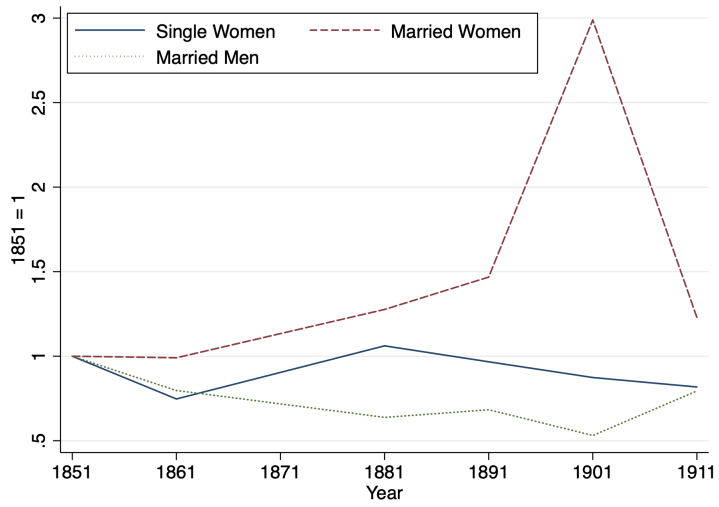
Sara Horrell and Jane Humphries. The origins and expansion of the male breadwinner family: The case of nineteenth-century britain. *International Review of Social History*, 42 (S5):25–64, 1997.

Chang-Tai Hsieh, Erik Hurst, Charles I Jones, and Peter J Klenow. The allocation of talent and us economic growth. *Econometrica*, 87(5):1439–1474, 2019.

MHD van Leeuwen, Ineke Maas, and Andrew Miles. *HISCO: Historical international standard classification of occupations*. Leuven: Leuven University Press, 2002.

Joel Mokyr. *The Enlightened economy an economic history of Britain 1700-1850*. Yale University Press, 2010.

Figure 8: Growth of misallocation



Notes: The figure plots the variance of $\frac{p_{ig}}{p_{i,sm}}$ over time, relative to 1851.

Andrew Donald Roy. Some thoughts on the distribution of earnings. *Oxford economic papers*, 3(2):135–146, 1951.

Pamela Sharpe. Continuity and change: women's history and economic history in britain. *The Economic History Review*, 48(2):353–369, 1995.

Daniela Vidart. Human Capital, Female Employment, and Electricity: Evidence from the early 20th-century United States. *Review of Economic Studies*, 91(1):560–594, 2024.

Matthew Woollard. The classification of occupations in the 1881 census of england and wales. *History and Computing*, 10(1-3):17–36, 1998.

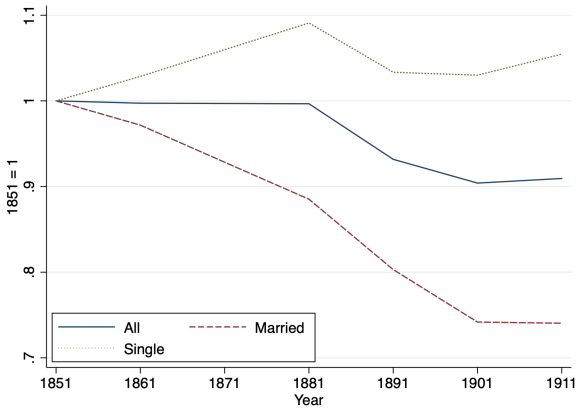
Xuesheng You. Women's labour force participation in nineteenth-century england and wales: evidence from the 1881 census enumerators' books. *The Economic History Review*, 73(1): 106–133, 2020a.

Xuesheng You. Working with husband? "occupation's wife" and married women's employment in the censuses in england and wales between 1851 and 1911. *Social Science History*, 44(4):585–613, 2020b.

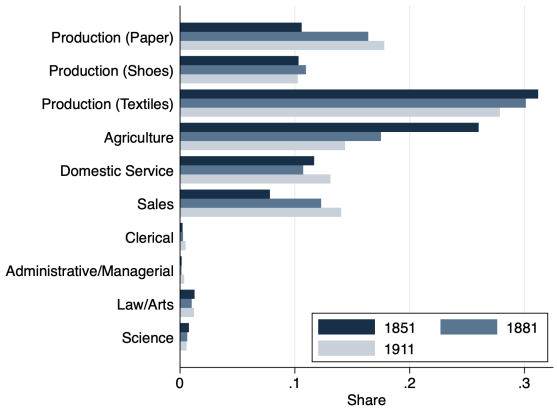
A Figures

Figure A.1: Robustness check: "Occupation's wife"

(a) Labor force participation



(b) Occupation distribution



Notes: Figure A.1a shows the labor force participation of women, assuming that 50% of all women married to men in the ten most common corresponding occupations associated with the term "occupation's wife" using the 'minor' HISCO codes in 1851 are in the labor force and work with their husbands. Labor force participation rate is normalized to be equal to 1 in 1851. Figure A.1b displays the resulting occupation distribution of married women.

B Tables

Table A.1: Labor force participation (1851-1911)

Year	1851	1861	1881	1891	1901	1911
Men						
Overall	88.8	87.8	89.1	89.1	92.3	89.2
Single	84.3	84.3	85.5	85.9	89.8	84.5
Married	92.2	90.5	91.9	91.6	94.3	93.0
Women						
Overall	41.3	41.8	42.1	39.0	37.6	38.6
Single	67.9	69.9	74.1	70.2	70.0	71.6
Married	18.0	18.5	15.5	12.1	9.7	11.4

Notes: The table reports the labor force participation of men and women, in levels.

Table A.2: Age-cohort regressions, individual level

Cohort	(1) Single	(2) Married	(3) Married
1787-96	-0.0175	0.0035	0.0132***
	(0.0104)	(0.0067)	(0.0019)
1807-16	-0.0026	-0.0114*	-0.0150***
	(0.0093)	(0.0058)	(0.0025)
1817-26	-0.0078	-0.0139*	-0.0275***
	(0.0113)	(0.0068)	(0.0041)
1827-36	-0.0376**	-0.0221***	-0.0433***
	(0.0152)	(0.0042)	(0.0046)
1837-46	-0.0262*	-0.0461***	-0.0668***
	(0.0151)	(0.0044)	(0.0066)
1847-56	-0.0085	-0.0630***	-0.0862***
	(0.0090)	(0.0048)	(0.0061)
1857-66	0.0014	-0.0777***	-0.1001***
	(0.0143)	(0.0054)	(0.0046)
1867-76	-0.0188	-0.0942***	-0.1204***
	(0.0156)	(0.0076)	(0.0050)
1877-86	-0.0128	-0.1001***	-0.1397***
	(0.0168)	(0.0086)	(0.0064)
1887-96	0.0034	-0.1070***	-0.1568***
	(0.0192)	(0.0129)	(0.0085)
Year Effect	No	No	Yes
Observations	19,130,225	25,004,732	44,134,957

Notes: The dependent variable is the labor force status of individual l in age group a at time t. Column (1) displays cohort-effects of single women and Column (2) and (3) considers those of married women. Standard errors are reported in parentheses and are clustered at the cohort level. The excluded cohort is 1797-1806. All regressions include a full set of age effects (10-year age bands). *** p<0.01, ** p<0.05, * p<0.1

C Model appendix

Proposition 1. Individuals choose their occupation to maximize lifetime utility. The probability of choosing a given occupation is then:

$$\Pr(i = \max U_j) = \Pr(U_i \geq \max U_j)$$

Let $U_i^{\frac{1}{3}} = \bar{z}_{ig}(c)[1 - \tau_{ig}^w(c)]w_i(c)\bar{\gamma}\bar{h}_{ig}\epsilon_i = \tilde{w}_{ig}\epsilon_i$

The following will be helpful in deriving the probability that i is the maximum:

$$\begin{aligned}\Pr(U_i^{\frac{1}{3}} \leq x) &= \Pr\left(\epsilon_i \leq \frac{x}{\tilde{w}_{ig}}\right) \\ &= \exp\left(-\frac{x}{\tilde{w}_{ig}}^{-\theta}\right)\end{aligned}$$

Taking the derivative of this gives the pdf:

$$\Pr(U_i^{\frac{1}{3}} = x) = \theta\left(\frac{x}{\tilde{w}_{ig}}\right)^{-\theta} x^{-1}\exp^{-\frac{x}{\tilde{w}_{ig}}^{-\theta}}$$

The probability that i is the maximum can then be derived as follows:

$$\begin{aligned}\Pr(U_i \geq \max U_j) &= \int_0^\infty \Pr(U_i^{\frac{1}{3}} = x)\prod_{j=1}^{M}\Pr(U_i^{\frac{1}{3}} \geq U_j^{\frac{1}{3}})dx \\ &= \int_0^1 \theta\left(\frac{x}{\tilde{w}_{ig}}\right)^{-\theta} x^{-1}\exp\left(-\sum_{j=1}^{I}\frac{x}{\tilde{w}_{ig}}^{-\theta}\right)dx \\ &= \frac{\tilde{w}_{ig}^\theta}{\sum_{j=1}^I \tilde{w}_{jg}^\theta}\int_0^\infty \theta\sum_{j=1}^{I}\tilde{w}_{jg}^\theta x^{-\theta-1}\exp\left(-\sum_{j=1}^{I}\tilde{w}_{jg}^\theta x^{-\theta}\right)dx \\ &= \frac{\tilde{w}_{ig}^\theta}{\sum_{j=1}^I \tilde{w}_{jg}^\theta}\end{aligned}$$

Where the last inequality follows because the term inside the integral is the pdf of the following Frechet distribution: $F(x) = \exp\left(-Mx^\theta\right)$, where $M = \sum_{j=1}^I \tilde{w}_{jg}^\theta$. Thus, integrating over its entire support equals one. □

Proposition 2. As noted in the proposition statement,

$$\overline{wage}_{ig}(c,t) = [1 - \tau_{ig}^w(t)]w_i(t)\mathbb{E}\left[\gamma(t-c)\bar{h}_{ig}\epsilon_i|\text{chooses i}\right]$$

.

First note that:

$$\mathbb{E}\left[\gamma(t-c)\bar{h}_{ig}\epsilon_i|\text{chooses i}\right] = \gamma(t-c)\bar{h}_{ig}\mathbb{E}\left[\epsilon_i|\text{chooses i}\right]$$

.

First, let's derive the relevant probability distribution for $\Pr(\epsilon_i \leq x|choosesi)$. Let F denote the CDF of ϵ_i and f denote its pdf:

$$\begin{aligned}
G_i(x) \equiv \Pr(\epsilon_i \leq x|choosesi) &= \Pr(\epsilon_i \leq x|\tilde{w}_{ig}\epsilon_i \geq \tilde{w}_{jg}\epsilon_j \forall j) \\
&= \frac{\Pr(\epsilon \leq x, \tilde{w}_{ig}\epsilon_i \geq \tilde{w}_{jg}\epsilon_j \forall j)}{\Pr(\tilde{w}_{ig}\epsilon_i \geq \tilde{w}_{jg}\epsilon_j \forall j)} \\
&= \frac{\int_0^x f(x)\prod_{j=1}^M F\left(\frac{\tilde{w}_i}{\tilde{w}_j}x\right)dx}{\frac{\tilde{w}_{ig}^\theta}{\sum_{j=1}^I \tilde{w}_{jg}^\theta}} \\
&= \int_0^x \theta x^{-\theta-1}\left(\sum_{j=1}^I \left(\frac{\tilde{w}_{jg}}{\tilde{w}_{ig}}\right)^\theta\right)\exp\left(-x^{-\theta}\left(\sum_{j=1}^I \left(\frac{\tilde{w}_{jg}}{\tilde{w}_{ig}}\right)^\theta\right)\right)dx
\end{aligned}$$

Then:

$$\begin{aligned}
\mathbb{E}\left[\epsilon_i|\text{chooses i}\right] &= \int_0^\infty z g_i(x)dx \\
&= \int_0^\infty z\theta x^{-\theta-1}\sum_{j=1}^M \left(\frac{\tilde{w}_{jg}}{\tilde{w}_{ig}}\right)^\theta \exp\left(-x^{-\theta}\left(\sum_{j=1}^M \frac{\tilde{w}_{jg}}{\tilde{w}_{ig}}^\theta\right)\right)dx \\
&= \left(\sum_{j=1}^I \left(\frac{\tilde{w}_{jg}}{\tilde{w}_{ig}}\right)^\theta\right)^{\frac{1}{\theta}} \int_0^\infty y^{-\frac{1}{\theta}}e^{-y}dy \\
&= p_{ig}^{-\frac{1}{\theta}}\Gamma\left(\frac{\theta-1}{\theta}\right)
\end{aligned}$$

Where the third inequality uses a change of variable, $y = z^{-\theta}\sum_{j=1}^I \left(\frac{\tilde{w}_{jg}}{\tilde{w}_{ig}}\right)^\theta$, and $\Gamma\left(\frac{\theta-1}{\theta}\right)$ is the gamma function evaluated at $\frac{\theta-1}{\theta}$. The remaining steps of the proof are to plug this back into the overall average wage function, and then to replace $\tilde{w}_{ig}$ with its definition and re-arrange. □

Proposition 3/4. Plug in definitions from Proposition 1/2 and re-arrange. □

D Data appendix

This section describes the two-digit HISCO occupations contained in each of the nine occupation groups.

0. SCIENCE - N
    - Physical Scientists and Related Technicians - MC
    - Architects, Engineers and Related Technicians - MC
    - Aircraft and Ships' Officers - MC
    - Life Scientists and Related Technicians - MC
    - Medical, Dental, Veterinary and Related Workers - MC
    - Statisticians, Mathematicians, Systems Analysts and Related Technicians - MC
    - Economists - MC
1. LAW AND ARTS
    - Accountants - MC; ; N
    - Jurists - unsure; N
    - Teachers - MC or LC; N
    - Workers in Religion - MC; N
    - Authors, Journalists and Related Writers - MC or LC; ; N
    - Sculptors, Painters, Photographers and Related Creative Artists - MC or LC; N
    - Composers and Performing Artists - MC or LC; N
    - Athletes, Sportsmen and Related Workers - MC; N
    - Professional, Technical and Related Workers Not Elsewhere Classified - MC; N
2. ADMINISTRATIVE AND MANAGERIAL WORKERS
    - Legislative Officials and Government Administrators - MC; N
    - Managers - MC; N
    - Supervisors, Foremen and Inspectors - MC; N
3. CLERICAL AND RELATED WORKERS; N

- Clerical and Related Workers, Specialisation Unknown MC
- Government Executive Officials MC
- Stenographers, Typists and Card- and Tape-Punching Machine Operators MC
- Bookkeepers, Cashiers and Related Workers MC
- Computing Machine Operators MC
- Transport Conductors MC
- Mail Distribution Clerks MC
- Telephone and Telegraph Operators MC
- Clerical and Related Workers Not Elsewhere Classified MC

4. SALES WORKERS LC MC;
   - Working Proprietors (Wholesale and Retail Trade) – potentially
   - Buyers
   - Technical Salesmen, Commercial Travellers and Manufacturers Agents
   - Insurance Real Estate, Securities and Business Services Salesmen and Auctioneers
   - Sales Workers Not Elsewhere Classified
5. DOMESTIC SERVICE WORKERS LC
   - Working Proprietors (Catering, Lodging and Leisure Services); T
   - Cooks, Waiters, Bartenders and Related Workers; T- to look into
   - Maids and Related Housekeeping Service Workers Not Elsewhere Classified; N
   - Building Caretakers, Charworkers, Cleaners and Related Workers; N
   - Launderers, Dry-Cleaners and Pressers; T
   - Hairdressers, Barbers, Beauticians and Related Workers; T
   - Protective Service Workers; N
   - Service Workers Not Elsewhere Classified; T
6. AGRICULTURE LC - T
   - Farmers
   - Agricultural and Animal Husbandry Workers

- Forestry Workers
- Fishermen, Hunters and Related Workers

7. PRODUCTION (Textiles)
   - Miners, Quarrymen, Well Drillers and Related Workers LC; N
   - Metal Processors LC; N
   - Wood Preparation Workers and Paper Makers LC; N
   - Chemical Processors and Related Workers L; N
   - Spinners, Weavers, Knitters, Dyers and Related Workers lC; T
   - Tanners, Fellmongers and Pelt Dressers LC; T
   - Food and Beverage Processors LC; T
   - Tobacco Preparers and Tobacco Product Makers LC; T
   - Tailors, Dressmakers, Sewers, Upholsterers and Related Workers LC; T
8. PRODUCTION (Shoes)
   - Shoemakers and Leather Goods Makers LC; T
   - Cabinetmakers and Related Woodworkers LC; T
   - Stone Cutters and Carvers LC; T
   - Blacksmiths, Toolmakers and Machine Tool Operators LC
   - Machinery Fitters, Machine Assemblers and Precision-Instrument Makers (except Electrical) MC; N
   - Electrical Fitters and Related Electrical and Electronics Workers MC; N
   - Broadcasting Station and Sound Equipment Operators and Cinema Projectionists MC; N
   - Plumbers, Welders, Sheet Metal and Structural Metal Preparers and Erectors MC; N
   - Jewellery and Precious Metal Workers MC; T
   - Glass Formers, Potters and Related Workers LC; T
9. PRODUCTION (Paper) LC; N
   - Rubber and Plastics Product Makers

- Paper and Paperboard Products Makers
- Printers and Related Workers
- Painters
- Production and Related Workers Not Elsewhere Classified
- Bricklayers, Carpenters and Other Construction Workers
- Stationary Engine and Related Equipment Operators
- Material Handling and Related Equipment Operators, Dockers and Freight Handlers
- Transport Equipment Operators